INCH-POUND

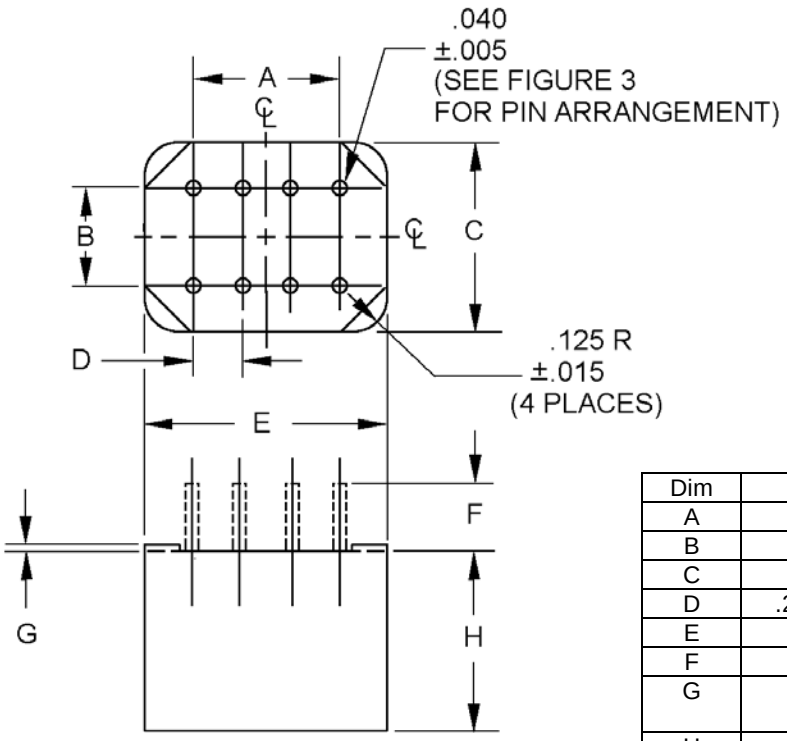
MIL-PRF-27/165E
w/AMENDMENT 1
1 November 2016
SUPERSEDING
MIL-PRF-27/165E
1 October 2010

PERFORMANCE SPECIFICATION SHEET

TRANSFORMER AUDIO FREQUENCY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-27.




CIRCUIT DIAGRAM AND MARKING
(SEE FIGURE 2 AND TABLE I)

| Dim | Inches | mm |
|---|---|---|
| A | .600 | 15.24 |
| B | .400 | 10.16 |
| C | .750 Max. | 19.05 Max. |
| D | .200 (3 places) | 5.08 (3 places) |
| E | 1.000 Max. | 25.40 Max. |
| F | .281 ±.031 | 7.14 ±.79 |
| G | .031 ±.015<br>(4 places) | .79 ±.38<br>(4 places) |
| H | .719 Max. | 18.26 Max. |

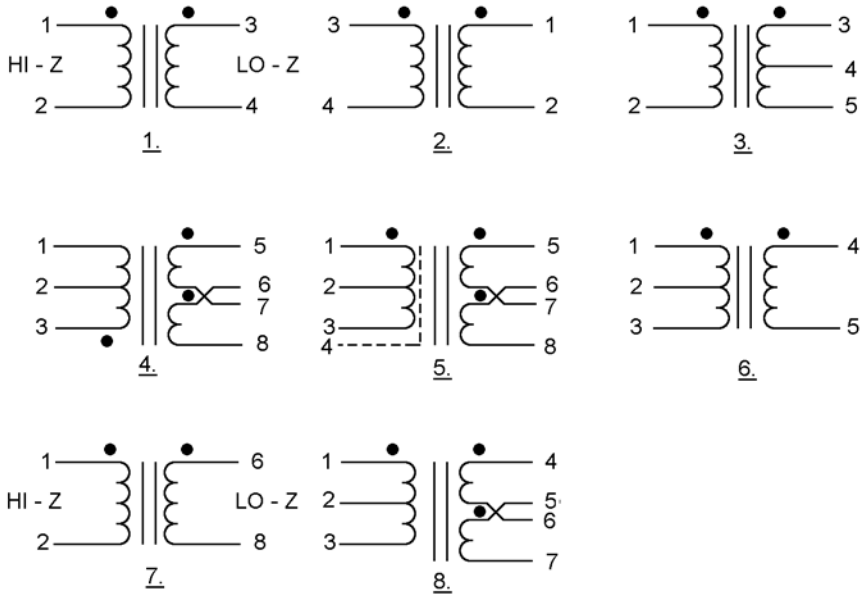
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side(s) and top of case, or both.
4. Unless otherwise specified, the tolerance shall be ±.010 (0.25 mm).

FIGURE 1. Dimensions and configurations.

AMSC N/A

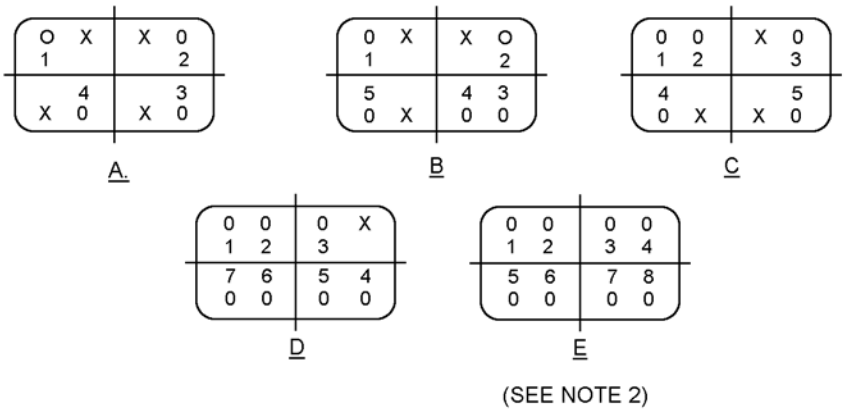
FSC 5950

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NOTE: Dots indicate the same polarity on terminals.

FIGURE 2. Circuit diagram.




(SEE NOTE 2)

NOTES

1. Pins not used are removed. These are indicated by “X”.
2. Pin numbers not shown in the circuit diagram (see figure 2) indicates that those pins will be missing.

FIGURE 3. Pin arrangement.



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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Working voltage: 175 volts.

Frequency range: See table II.

Duty cycle: Continuous.

Design and construction:

Dimensions and configurations: See figure 1.

Case: Plastic/molded.

Terminals: Pin type, .040 inch diameter (1.02 mm) (see figure 3 for pin arrangement)

Material: Type LW ( )C5N in accordance with MIL-STD-1276.

Weight: 1.0 ounce maximum.

Altitude: 75,000 feet maximum.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202-211, test condition A, 2.0 pounds.

Vibration (high frequency): MIL-STD-202-204.

Dielectric withstanding voltage: Between split sections, 50 volts rms maximum.

At sea level: 500 V rms.

At reduced barometric pressure: 300 V rms.

Electrical characteristics:

Frequency response: See table II

Reference frequency: 1kHz

No load (center tap voltage unbalance only) ±1 percent at 1 volt, 1 kHz across the primary.

Marking location: See figure 1.

Electrostatic shielding: Shielding ratio 5.1 minimum at 20 kHz applicable to circuit diagram 5 only.

Part or Identifying Number (PIN): M27/165-(dash number from table I).

NOTE: Where dual ratings are specified, only frequency response shall be tested in both modes of operation.



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TABLE I Electrical ratings.

| Dash 1/ no. | Primary 2/ impedance (ohms) | Secondary 2/ impedance (ohms) | Primary DCR ±25% (ohms) | Secondary DCR ±25% (ohms) | Power Level (mW) | Primary dc current (mA) | Pin arrangement (see figure 3) | Circuit diagram (see figure 2) |
|---|---|---|---|---|---|---|---|---|
| 01 | (1-3)<br>80 CT<br>100 CT | 32 SPLIT<br>40 SPLIT | 4.9 | 3.2 | 250 | 16 | D | 8 |
| | (4-7)<br>32 SPLIT<br>40 SPLIT | 80 CT<br>100 CT | 3.2 | 4.9 | 250 | 25 | D | 8 |
| 02 | (6-8)<br>200 | 250 k | 16 | 2500 | 10 | 0 | E | 7 |
| | (1-4)<br>250 k | 200 | 2500 | 16 | 10 | 0 | E | 7 |
| 03 | (1-3)<br>400 CT<br>500 CT | 40 SPLIT<br>50 SPLIT | 20 | 4.5 | 250 | 8 | E | 4 |
| | (5-8)<br>40 SPLIT<br>50 SPLIT | 400 CT<br>500 CT | 4.5 | 20 | 250 | 25 | E | 4 |
| 04 | (1-3)<br>400 CT<br>500 CT | 120 SPLIT<br>150 SPLIT | 20 | 12.5 | 250 | 8 | E | 4 |
| | (5-8)<br>120 SPLIT<br>150 SPLIT | 400 CT<br>500 CT | 12.5 | 20 | 250 | 14 | E | 4 |
| 05 | (1-3)<br>400 CT<br>500 CT | 400 SPLIT<br>500 SPLIT | 20 | 45 | 250 | 8 | E | 4 |
| | (5-8)<br>400 SPLIT<br>500 SPLIT | 400 CT<br>500 CT | 45 | 20 | 250 | 8 | E | 4 |
| 06 | (1-3)<br>500 CT | 3.2 | 15 | 0.35 | 250 | 0 | C | 6 |
| | (4-5)<br>3.2 | 500 CT | 0.35 | 15 | 250 | 0 | C | 6 |
| 07 | (1-3)<br>600 CT | 600 SPLIT | 35 | 60 | 250 | 6 | D | 8 |
| | (4-7)<br>600 SPLIT | 600 CT | 60 | 35 | 250 | 6 | D | 8 |

See footnote at end of table.



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TABLE I Electrical ratings - Continued.

| Dash 1/ no. | Primary 2/ impedance (ohms) | Secondary 2/ impedance (ohms) | Primary DCR ±25% (ohms) | Secondary DCR ±25% (ohms) | Power Level (mW) | Primary dc current (mA) | Pin arrangement (see figure 3) | Circuit diagram (see figure 2) |
|---|---|---|---|---|---|---|---|---|
| 08 3/ | (5-8)<br>600 SPLIT | 10 k CT | 80 | 1050 | 200 | 9 | E | 5 |
| | (1-3)<br>10 k CT | 600 SPLIT | 1050 | 80 | 200 | 4 | E | 5 |
| 09 | (1-3)<br>2 k CT<br>4 k CT | 8<br>16 | 290 | 2 | 250 | 4<br>2 | C | 6 |
| | (4-5)<br>8<br>16 | 2 k CT<br>4 k CT | 2 | 290 | 250 | 60 | C | 6 |
| 10 | (1-4)<br>2.5 k CT | 2.5 k SPLIT | 140 | 300 | 250 | 4 | E | 4 |
| | (5-8)<br>2.5 k SPLIT | 2.5 k CT | 300 | 140 | 250 | 4 | E | 4 |
| 11 | (1-2)<br>10 k<br>25 k | 200<br>500 | 1225 | 30 | 200 | 3<br>1.5 | A | 1 |
| | (3-4)<br>200<br>500 | 10 k<br>25 k | 30 | 1225 | 200 | 21<br>10 | A | 1 |
| 12 | (1-2)<br>10 k | 2 k CT | 1000 | 40 | 200 | 1 | B | 3 |
| | (3-5)<br>2 k CT | 10 k | 40 | 1000 | 200 | 2.2 | B | 3 |
| 13 3/ | (1-3)<br>10 k CT<br>12 k CT | 10 k SPLIT<br>12 k SPLIT | 855 | 1080 | 200 | 1 | E | 5 |
| | (5-8)<br>10 k SPLIT<br>12 k SPLIT | 10 k CT<br>12 k CT | 1080 | 855 | 200 | 1 | E | 5 |
| 14 | (3-4)<br>10 k | 90 k | 215 | 1850 | 100 | 0.25 | A | 2 |
| | (1-2)<br>90 k | 10 k | 1850 | 215 | 100 | 0 | A | 2 |

See footnote at end of table.



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TABLE I Electrical ratings - Continued.

| Dash 1/ no. | Primary 2/ impedance (ohms) | Secondary 2/ impedance (ohms) | Primary DCR ±25% (ohms) | Secondary DCR ±25% (ohms) | Power Level (mW) | Primary dc current (mA) | Pin arrangement (see figure 3) | Circuit diagram (see figure 2) |
|---|---|---|---|---|---|---|---|---|
| 15 | (1-2)<br>20 k<br>30 k | <br>800<br>1200 | 450 | 32 | 200 | 0.5 | A | 1 |
| | (3-4)<br>800<br>1200 | <br>20 k<br>30 k | 32 | 450 | 200 | 2.5 | A | 1 |
| 16 | (1-4)<br>30 k | <br>50 | 1850 | 3.8 | 200 | 1 | E | 7 |
| | (6-8)<br>50 | <br>30 k | 3.8 | 1850 | 200 | 24 | E | 7 |
| 17 | (1-3)<br>50 k CT | <br>600 SPLIT | 2400 | 63 | 250 | 1 | D | 8 |
| | (4-7)<br>600 SPLIT | <br>50 k CT | 63 | 2400 | 250 | 9 | D | 8 |
| 18 | (1-4)<br>100 k | <br>60 | 3400 | 3.7 | 200 | 0.5 | E | 7 |
| | (6-8)<br>60 | <br>100 k | 3.7 | 3400 | 200 | 20 | E | 7 |

1/ Dual rating for each dash number indicates that the transformer's winding can be used arbitrarily as primary or secondary.
2/ Impedance values written one above the other indicates a range of matching impedances over Which the parts will give satisfactory performance as long as the impedance ratio is maintained.
3/ Electrostatic shield, shield ratio 5:1 minimum at 10 kHz.


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TABLE II Electrical characteristics.

| 1/ Dash number | Frequency response at 200 Hz to 20 kHz ±3 dB | | | Polarity same on terminals (see figure 2) |
|---|---|---|---|---|
| | $Z_S$ (ohms) | $Z_L$ (ohms) | $E_L$ (ohms) | |
| 01 | (1-3) 80<br>(4-7) 32 | 32<br>80 | 2.8<br>4.4 | 1, 4, 5<br>1, 4, 5 |
| 02 | (6-8) 200<br>(1-4) 250 k | 250 k<br>200 | 50<br>1.4 | 1, 6<br>1, 6 |
| 03 | (1-3) 400<br>(5-18) 40 | 40<br>400 | 3.16<br>10 | 3, 5, 6<br>3, 5, 6 |
| 04 | (1-3) 400<br>(5-8) 120 | 120<br>400 | 5.5<br>10 | 3, 5, 6<br>3, 5, 6 |
| 05 | (1-3) 400<br>(5-8) 400 | 400<br>400 | 10<br>10 | 3, 5, 6<br>3, 5, 6 |
| 06 | (1-3) 500<br>(4-5) 3.2 | 3.2<br>500 | 0.9<br>11 | 1, 4<br>1, 4 |
| 07 | (1-3) 600<br>(4-7) 600 | 600<br>600 | 12.2<br>12 | 1, 4, 5<br>1, 4, 5 |
| 08 | (5-8) 600<br>(1-3) 100 k | 10 k<br>600 | 45<br>12 | 1, 5, 6<br>1, 5, 6 |
| 09 | (1-3) 2 k<br>(4-5) 8 | 8<br>2 k | 1.4<br>22 | 1, 4<br>1, 4 |
| 10 | (1-4) 2.5 k<br>(5-8) 2.5 k | 2.5 k<br>2.5 k | 25<br>25 | 3, 5, 6<br>3, 5, 6 |
| 11 | (1-2) 10 k<br>(3-4) 200 | 200<br>10 k | 6.3<br>40 | 1, 3<br>1, 3 |
| 12 | (1-2) 10 k<br>(3-5) 2 k | 2 k<br>10 k | 20<br>40 | 1, 3<br>1, 3 |
| 13 | (1-3) 10 k<br>(5-8) 10 k | 10 k<br>10 k | 45<br>45 | 1, 5, 6<br>1, 5, 6 |
| 14 | (3-4) 10 k<br>(1-2) 90 k | 90 k<br>10 k | 95<br>30 | 1, 3<br>1, 3 |
| 15 | (1-2) 20 k<br>(3-4) 800 | 800<br>20 k | 12.6<br>60 | 1, 3<br>1, 3 |
| 16 | (1-4) 30 k<br>(6-8) 50 | 50<br>30 k | 3.16<br>77 | 1, 6<br>1, 6 |
| 17 | (1-3) 50 k<br>(4-7) 600 | 600<br>50 k | 12.2<br>110 | 1, 4, 5<br>1, 4, 5 |
| 18 | (1-4) 100 k<br>(6-8) 60 | 60<br>100 k | 3.4<br>170 | 1, 6<br>1, 6 |

1/ Qualification testing and approval to M27/165-12 shall be sufficient to grant qualification approval to M27/165-01 through M27/165-18.



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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204 MIL-STD-202-211 MIL-STD-1276

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2016-029)

Review activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

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